

TREPP: Tandem Repeat Expansion Pathogenicity Prediction via Stacked CatBoost and Context-Aware Sequence Features

Minghua Xu^{1,3}, Kang Hu^{1,2,3}, Chao Deng^{1,3}, Peng Ni^{1,2,3}, and Jianxin Wang^{1,2,3}

¹School of Computer Science and Engineering, Central South University, Changsha, 410083, China

²Xiangjiang Laboratory, Changsha, 410205, China

³Hunan Provincial Key Lab on Bioinformatics, Central South University, Changsha, 410083, China

Email: jxwang@mail.csu.edu.cn

Abstract—Tandem repeat (TR) expansions are implicated in a broad range of human disorders. However, predicting their pathogenic potential remains challenging due to the scarcity of validated loci, extreme class imbalance, and the dependence of repeat instability on complex local sequence context. Existing approaches primarily rely on gene-centric association signals and often overlook intrinsic sequence features that contribute to repeat instability, such as local repeat density and differences in GC content between the repeat and its flanking regions.

We introduce TREPP, an interpretable ensemble learning framework designed to robustly assess the pathogenic potential of TR loci. TREPP combines genomic location with sequence-based features to capture the sequence context of expansion. To address label scarcity, the framework employs a multi-sampling strategy that trains multiple Optuna-optimized CatBoost base learners on diversified negative subsets. Subsequently, their calibrated outputs are integrated via a sparse logistic regression meta-learner.

Benchmarking on a manually curated dataset, TREPP achieves an AUPRC of 94.71%, outperforming the state-of-the-art method REXPERT by 6.11 percentage points. SHAP-based interpretability analysis confirms that the model leverages biologically plausible signals, while genome-wide application reveals a strong linear association between TREPP scores and enrichment in functional genomic regions. Furthermore, top-K prioritization analyses demonstrate that TREPP yields stronger enrichment of known disease-associated genes (OMIM and GeneTrek) than existing baselines. TREPP serves as a robust tool for identifying high-risk TR loci and is publicly available at <https://github.com/minghuaxu/TREPP>.

Index Terms—Tandem Repeat, Pathogenicity Prediction, CatBoost, Ensemble Learning.

I. INTRODUCTION

The human genome contains over one million tandem repeat (TR) loci, collectively accounting for approximately 3% of the total sequence [1], [2]. These repetitive elements are broadly categorized into short tandem repeats (STRs), with motif lengths of 1–6 base pairs, and variable number tandem repeats (VNTRs), with motifs exceeding 6 base pairs [3], [4]. Compared to single nucleotide polymorphisms (SNPs), TRs exhibit exceptionally high mutation rates [5], primarily driven by replication slippage that accelerates exponentially once the repeat copy number surpasses a critical threshold [6]. Beyond simple tract length, this instability is governed by the local biophysical environment: sequence motifs can fold into non-canonical “non-B” DNA conformations (e.g., hairpins and G-quadruplexes) that stall replication and promote expansion [7].

Crucially, the formation of these structures is modulated by the flanking sequence context, such as GC content, which can stabilize secondary structures and affect locus-specific instability [8]. To date, unstable TR expansions have been identified as the cause of over 80 nervous and neuromuscular diseases [7], [9]. These disorders arise from diverse pathogenic mechanisms, including loss of function, gain of function, and dominant-negative effects. Their clinical presentation depends on the expansion’s genomic location (e.g., coding vs. non-coding regions) and is complicated by features like anticipation and somatic mosaicism [8], [10], [10].

However, TR expansions remain understudied because of persistent technical challenges. Short-read sequencing often lacks the power to resolve long, complex repeat tracts or repeat-rich neighborhoods, hindering the identification of high-confidence disease associations [8], [11]. While recent large-scale profiling efforts have expanded the catalog of TRs [12], the number of experimentally validated pathogenic loci remains disproportionately small compared to the vast space of polymorphic TRs. For instance, the STRchive database currently contains approximately 75 curated TR loci with established disease phenotypes [13]. This scarcity creates a severe class imbalance in training datasets. Standard machine learning algorithms often perform suboptimally in such settings, as they tend to bias predictions toward the majority (benign) class to maximize overall accuracy [14]. A model that ignores these biologically informative sequence signals and class-imbalance issues may fail to generalize to understudied or novel loci.

To address these challenges, machine learning techniques—particularly ensemble methods like Boosting—have been widely applied to genetic disorder prognosis [15], [16]. However, dedicated computational tools for TR pathogenicity prediction remain scarce. To date, the primary tool in this niche is REXPERT [17], which classifies TRs using genome annotations such as constraint scores and transcription factor binding sites. Despite its pioneering contribution, REXPERT has two main limitations. First, it relies predominantly on external annotation features, neglecting intrinsic genomic context—specifically the GC content difference and motif interruptions—that mechanistically modulates repeat instability [8]. Second, it lacks specialized strategies to handle the extreme class imbalance inherent to TR datasets.

We introduce TREPP, an ensemble learning framework

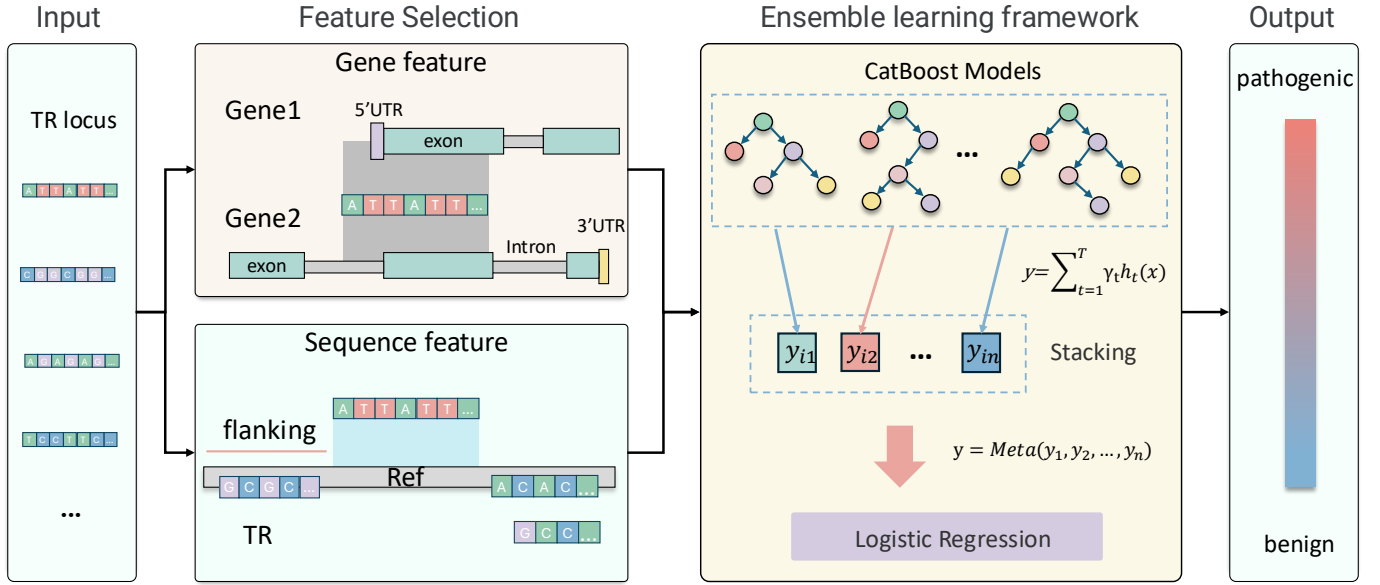


Fig. 1. **Overview of the TREPP framework.** The workflow consists of two main stages: feature extraction and ensemble learning. First, genomic features are extracted for each TR locus, including Gene features (spatial relationships to exons, introns, and UTRs) and Sequence features (GC content of the TR and its flanking region). Second, these features are concatenated into a unified vector and fed into an ensemble learning framework. Multiple CatBoost base learners generate initial predictions, which are then aggregated by a Logistic Regression meta-learner (Stacking strategy) to produce the final probability of pathogenicity.

designed to predict the pathogenic potential of TR loci with high accuracy and robustness. TREPP addresses these gaps in two ways. First, in addition to standard gene-level features, it explicitly incorporates sequence-context descriptors derived from the TR locus and its flanking regions, capturing compositional signals mechanistically linked to instability. Second, to mitigate class imbalance, TREPP employs a multi-model ensemble strategy: it trains multiple CatBoost base learners on diverse, balanced subsets of negative samples and aggregates their predictions via a meta-learner. This design enhances model diversity and sensitivity under positive scarcity. Benchmarking results demonstrate that TREPP significantly outperforms RExPRT, offering a powerful tool for identifying high-risk pathogenic TRs.

II. METHODS

Figure 1 illustrates the overall workflow of TREPP. The framework integrates both gene-level and sequence-level features to characterize each candidate TR locus. These features are combined into a unified vector and fed into an ensemble model. Specifically, multiple CatBoost base learners perform the initial classification. Their outputs are then integrated using a stacking strategy, serving as input features for a logistic regression meta-learner. Finally, the meta-learner calculates the pathogenicity probability, generating a binary prediction (0 for benign, 1 for pathogenic).

A. Feature Extraction

TREPP represents each candidate TR locus using a unified feature vector that captures: (i) the spatial relationships between TRs and gene structures, and (ii) repeat-intrinsic properties along with neighborhood sequence context. All features

are computed based on the reference genome and standard gene annotations. The same feature-engineering pipeline is applied consistently during both training and inference.

1) *TR-Gene Positional and Distance Features:* We denote a TR locus by $TR_i = (c, s_i, e_i)$, where c is the chromosome and $[s_i, e_i]$ is the genomic interval. We quantify its positional relationship to gene structures using GENCODE annotations (release v49lift37) [18]. For each locus, we compute the minimum distance to the nearest exon, coding sequence (CDS), and untranslated region (UTR), denoted as $dist_to_exon$, $dist_to_cds$, and $dist_to_utr$, respectively (distance is set to 0 if the TR overlaps the corresponding element). We also derive a gene-level positional descriptor by associating each TR with an overlapping gene (if available) or the nearest gene within a bounded neighborhood. We then compute the distance from the TR center to the transcription start site (TSS), $dist_to_tss$, along with the gene length ($gene_len$) and the relative position of the TR within the gene body ($gene_rel_pos \in [0, 1]$), accounting for gene strand orientation. To reduce the influence of outliers, distance features are clipped to an upper bound of 50 kb.

To improve interpretability, we further derive binary indicators, including in_gene (whether a gene assignment exists), in_exon and in_cds (whether the TR overlaps an exon or CDS), and promoter-proximity indicators such as $near_tss_1k$ and $near_tss_5k$ (indicating if $dist_to_tss \leq 1$ kb or 5 kb, respectively). Additionally, we normalize the TSS distance by gene length:

$$norm_dist_tss = \frac{dist_to_tss}{\max(gene_len, 1)}. \quad (1)$$

2) *Repeat-Intrinsic and Neighborhood Context Features:* We characterize intrinsic repeat properties and local genomic

context using three complementary components. First, we compute repeat length ($\text{tr_len} = e_i - s_i$), motif length (motif_len), and the estimated number of repeat units:

$$\text{repeat_units} = \frac{\text{tr_len}}{\max(\text{motif_len}, 1)}. \quad (2)$$

Second, we compute the GC content for the TR core (tr_gc) and for a local region spanning the TR with ± 1 kb flanking sequences (flank_gc_1k). We also explicitly capture the difference between the repeat and its neighborhood by calculating the absolute GC difference:

$$|\Delta \text{GC}_{\text{tr,flank}}| = |\text{tr_gc} - \text{flank_gc_1k}|. \quad (3)$$

Third, using Tandem Repeats Finder (TRF v4.10.0) [19], we quantify the local repeat density by calculating the fraction of bases covered by other repeats within two radii (1 kb and 10 kb), producing str_density_1000 and str_density_10000 . We also include a scale-density ratio to measure local variability:

$$\text{str_density_ratio} = \frac{\text{str_density_1000}}{\text{str_density_10000} + \epsilon}, \quad (4)$$

where ϵ is a small constant for numerical stability.

3) *Motif Composition and Unified Engineering*: To capture biochemical properties of the repeat unit, we compute motif-level descriptors, including GC/AT fractions (motif_gc , motif_at), purine/pyrimidine fractions, and a CpG indicator (motif_has_cpg). We also include motif-type indicators (e.g., $\text{is_trinuc} = 1$ if $\text{motif_len} = 3$).

Finally, we apply a unified feature-engineering module. We apply log-transformations to skewed numerical features (e.g., $\log_{10}(\text{tr_len} + 1)$, $\log_{10}(\text{repeat_units} + 10^{-3})$) and log-distance transforms to positional features. We also include features measuring the difference between TR GC content and motif GC content. The resulting feature vector provides a compact representation that integrates gene-level signals with repeat-intrinsic and neighborhood-context properties at multiple scales.

B. Model Training

Following feature extraction, TREPP employs a stacked ensemble framework designed to mitigate extreme class imbalance and enhance robustness given the scarcity of validated pathogenic loci. The architecture consists of (i) multiple CatBoost base learners trained on diversified undersampled subsets, and (ii) an L1-regularized logistic regression meta-learner that integrates calibrated base probabilities into a final pathogenicity score.

1) *Diversified Base Learners via Controlled Undersampling*: Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote the training set, where $y_i \in \{0, 1\}$ represents the benign or pathogenic class label. To ensure diversity among base models, we employ a controlled undersampling strategy. For each of the n base learners, we form a training subset containing all positive samples and a random $2 \times$ subset of negatives. This ensures each base model sees the full set of positives but a different subset of negatives, promoting diversity and reducing majority-class bias.

Each base learner is implemented using CatBoost, a gradient boosting decision tree algorithm optimized for heterogeneous

tabular data. We enable the model's internal class weighting parameters to further handle imbalance, and we initialize each learner with an independent random seed to encourage stochastic diversity.

2) *Hyperparameter Optimization for Base Models*: To maximize the performance of individual learners, we perform automated hyperparameter optimization using Optuna [20]. We tune parameters (including learning rate, tree depth, feature subsampling rate, and L2 regularization strength) by maximizing the mean Area Under the Precision–Recall Curve (AUPRC) over stratified 5-fold cross-validation. Early stopping prevents overfitting during optimization.

3) *Probability Calibration and Out-of-Fold (OOF) Construction*: For stacking, we need well-calibrated probabilities. Thus, we calibrate each optimized CatBoost model using Platt scaling (sigmoid calibration) via `CalibratedClassifierCV`. A critical step in stacking is the construction of an unbiased meta-feature matrix. For each base learner k , we generate a full-length Out-of-Fold (OOF) probability vector $\hat{p}^{(k)} \in [0, 1]^N$ covering the entire training set. Specifically:

- For samples included in the base learner's training subset, probabilities are generated via 5-fold cross-validation to prevent data leakage.
- For samples not included in the subset (i.e., the remaining majority class samples), probabilities are generated using the model trained on the full subset.

This procedure ensures that every entry in the meta-feature matrix is a prediction made by a model that did not see that specific sample during training. The final meta-feature matrix $Z \in \mathbb{R}^{N \times n}$ is formed by concatenating these vectors:

$$\mathbf{z}_i = [\hat{p}_i^{(1)}, \hat{p}_i^{(2)}, \dots, \hat{p}_i^{(n)}]. \quad (5)$$

4) *Meta-Learner*: TREPP employs logistic regression with L1 regularization as the meta-learner to aggregate the base models' outputs contained in Z . L1 regularization encourages sparsity, effectively selecting the most informative base models. To obtain an unbiased estimate of the ensemble's training performance, we compute the meta-level OOF probabilities using stratified 5-fold cross-validation on Z . Let $\hat{p}_i^{(\text{meta})}$ denote the resulting final predicted probability for sample i .

5) *Ensemble Size Determination*: The ensemble size n (number of base learners) was chosen empirically: we evaluated various n by CV AUROC and selected $n = 8$ as it maximized performance. AUROC was chosen as the selection criterion because it reflects the global ranking quality of the pathogenicity score, independent of a specific decision threshold.

III. RESULTS

A. Performance Evaluation

The dataset used in this study is derived from the manually curated REXPERT dataset. It defines pathogenic loci as those with fully validated causal relationships to diseases, while negative training data consists of TR loci that are commonly expanded (length > 150 bp) in healthy controls from the

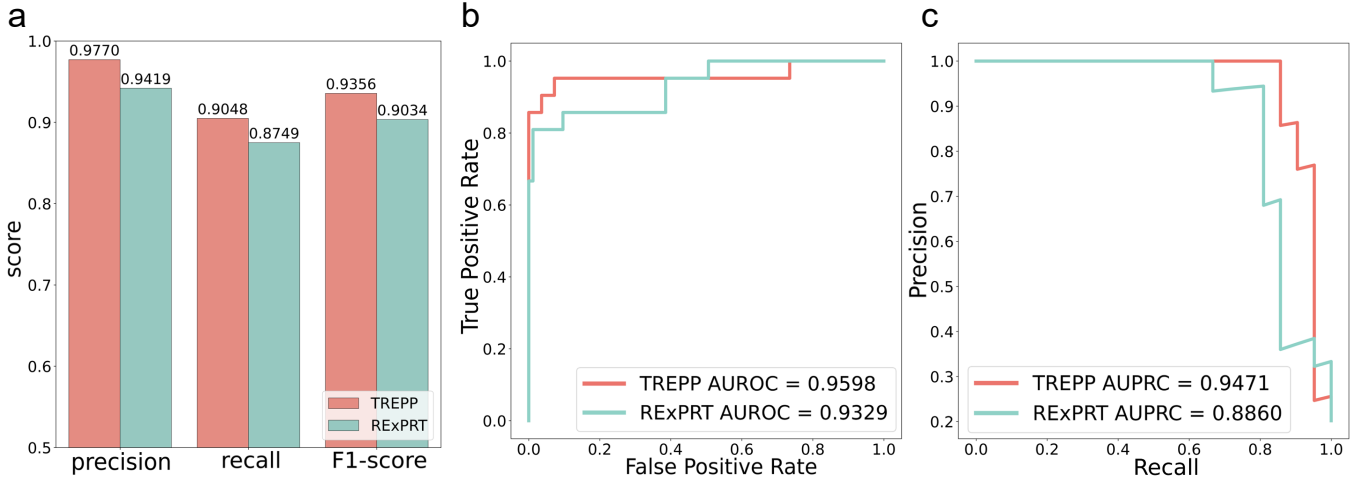


Fig. 2. **Performance comparison between TREPP and RExPRT.** (a) Comparison of threshold-dependent metrics (Precision, Recall, and F1-score) on the held-out test set at a fixed threshold of 0.5. (b) Receiver Operating Characteristic (ROC) curves showing the trade-off between sensitivity and specificity. (c) Precision-Recall (PR) curves highlighting performance under class imbalance. TREPP demonstrates consistently higher performance across all metrics compared to RExPRT.

1000 Genomes Project [21]. The final dataset includes 60 pathogenic TR loci and 700 benign control samples, presenting a significant class imbalance. To ensure a fair and rigorous comparison, we followed the exact same training and test split strategy as utilized in the RExPRT study. Specifically, the independent test set comprises 20 pathogenic samples and 83 benign controls, with the remaining samples allocated for training.

Compared to the state-of-the-art method RExPRT, TREPP demonstrates clear and consistent improvements across multiple evaluation metrics, demonstrating superior capability in handling class imbalance. As illustrated in Fig. 2a, at a standard decision threshold of 0.5, TREPP achieves a precision of 0.9770, a recall of 0.9048, and an F1-score of 0.9356. These results represent substantial performance gains of 3.51%, 2.99%, and 3.22% over RExPRT, respectively.

Given the substantial imbalance between positive and negative samples in the dataset, threshold-independent metrics provide a more comprehensive assessment of model robustness. As shown in the ROC and PR curves (Fig. 2b and c), TREPP achieves an Area Under the ROC Curve (AUROC) of 0.9598 and an Area Under the Precision-Recall Curve (AUPRC) of 0.9471. Notably, TREPP outperforms RExPRT by margins of 2.69% in AUROC and 6.11% in AUPRC, highlighting its enhanced sensitivity and precision in identifying rare pathogenic events.

Furthermore, TREPP effectively mitigates the bias typically caused by class imbalance through its specialized ensemble learning strategy. By training multiple base models on balanced subsets, TREPP learns to identify minority class samples more effectively than single-model approaches like RExPRT. This leads to higher recall without sacrificing precision. Importantly, TREPP shows improved detection of pathogenic TR loci located in intronic regions, a genomic context where existing methods often struggle due to complex sequence structures. By integrating both gene-level and sequence-level features, TREPP maintains robust predictive performance even

in these challenging non-coding regions.

B. Feature Interpretability Analysis

To evaluate the biological relevance of the features engineered in TREPP, we first analyzed the data distribution to determine if the proposed features could effectively distinguish pathogenic loci from benign ones. Figure 3a visualizes the feature space using Principal Component Analysis (PCA) and t-Distributed Stochastic Neighbor Embedding (t-SNE). In both projections, pathogenic loci do not scatter randomly; instead, they form clusters that are distinct from the benign loci. This separation indicates that our multi-dimensional feature set successfully captures the intrinsic differences between high-risk and benign repeats. However, the separation is not perfectly linear, and some overlap remains, reflecting the inherent complexity of TR pathogenicity. This observation justifies our use of non-linear ensemble models to learn these complex decision boundaries.

We further analyzed the distributions of specific key features using violin plots (Figure 3b). Positional features exhibited the most significant divergence between classes. Pathogenic loci are distributed much closer to functional gene elements, such as exons and coding sequences (indicated by lower values in `dist_to_exon` and `dist_to_cds`). This confirms the biological intuition that expansions located near or within coding regions carry a significantly higher risk of pathogenicity.

Sequence-based features also displayed clear distributional differences. Pathogenic loci tend to possess higher GC content (`log_tr_gc`) compared to benign controls. High GC content is known to facilitate the formation of non-B DNA structures, such as G-quadruplexes, which can stall replication forks and promote instability. Additionally, we observed a distinct pattern in the feature `delta_gc_tr_motif`. This feature measures the difference in GC content between the full repeat tract and the canonical repeat unit. Benign loci typically show very little difference (centered at zero), whereas pathogenic loci exhibit

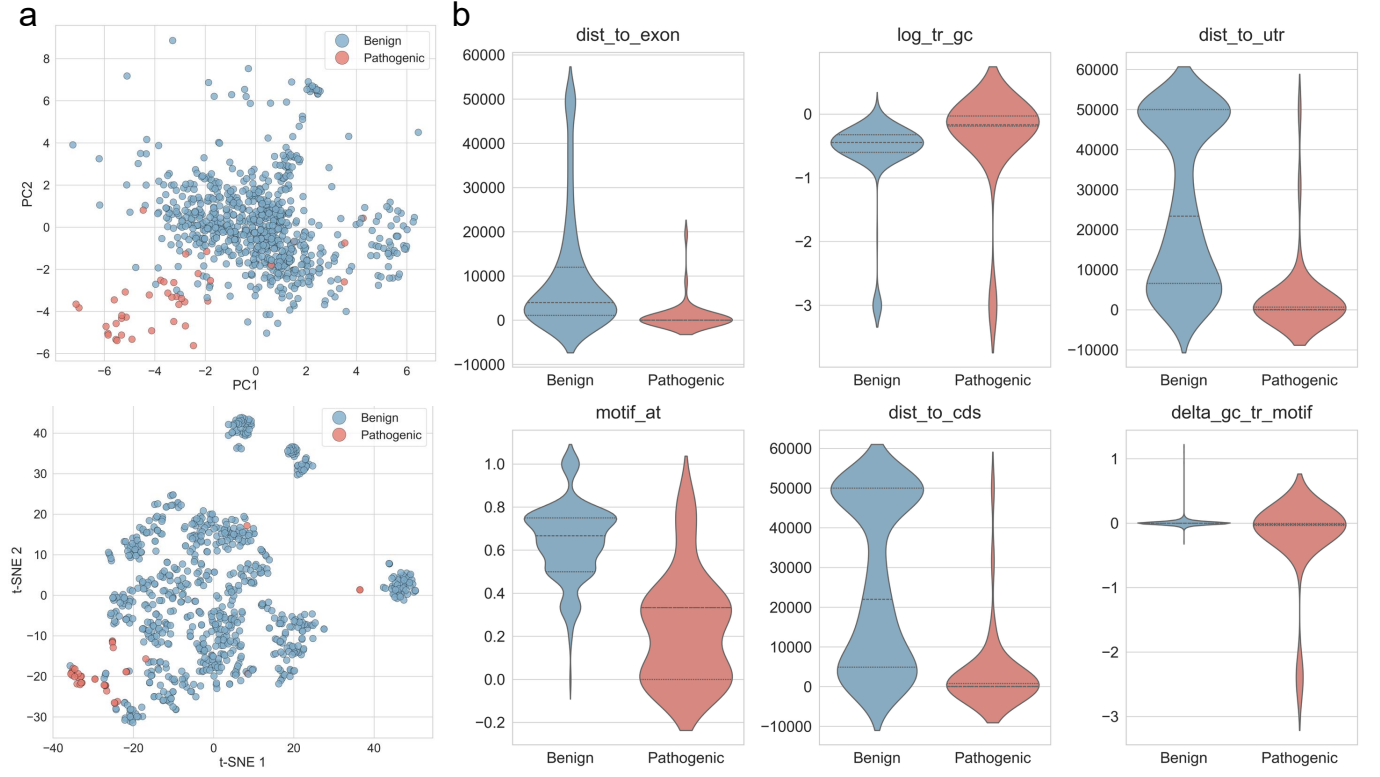


Fig. 3. **Feature interpretability analysis.** (a) PCA and t-SNE projections of TR loci using the extracted features. The plots show that pathogenic and benign loci tend to cluster in different regions, though some overlap remains. (b) Violin plots comparing key features between benign and pathogenic loci. Pathogenic loci show clear differences in their distance to gene elements and sequence composition.

a wider distribution. This suggests that high-risk loci often contain sequence interruptions or motif variants that deviate from the standard unit. In summary, these analyses confirm that our features are biologically interpretable and capture signals from both genomic location and sequence properties.

C. Ablation Experiment

To quantify the individual contributions of gene-related and sequence-derived features, we performed a comprehensive ablation study using three feature configurations: All (the full feature set), All-gene (excluding gene-related features), and All-seq (excluding sequence-based features). As shown in

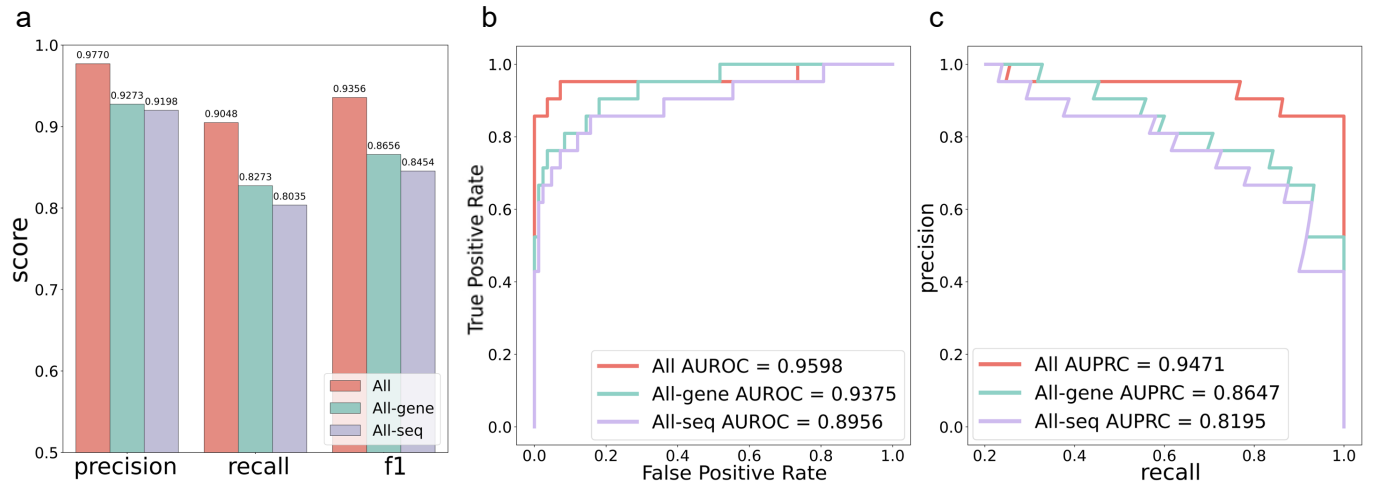


Fig. 4. **Ablation study of TREPP feature contributions.** (a) Comparison of threshold-dependent metrics (Precision, Recall, and F1-score) across different feature configurations at a fixed threshold of 0.5. (b) Receiver Operating Characteristic (ROC) curves showing the trade-off between True Positive Rate and False Positive Rate. (c) Precision-Recall (PR) curves showing model performance across varying thresholds. The three configurations evaluated are: All (full feature set), All-gene (ablation of gene-related features, retaining only sequence features), and All-seq (ablation of sequence-based features, retaining only gene features). AUROC and AUPRC values are annotated in the legends.

Fig. 4a, removing either feature group consistently degrades performance at the 0.5 threshold.

Specifically, when gene-related features are excluded (All-gene), the model's precision decreases from 0.9770 to 0.9273 (a drop of 4.97 percentage points), recall decreases from 0.9048 to 0.8273 (7.75 percentage points), and the F1-score decreases from 0.9356 to 0.8656 (7.00 percentage points). When sequence-based features are excluded (All-seq), the performance decline is even more pronounced, particularly in terms of sensitivity. In this configuration, precision drops from 0.9770 to 0.9198 (5.72 percentage points), recall drops significantly from 0.9048 to 0.8035 (10.13 percentage points), and the F1-score drops from 0.9356 to 0.8454 (9.02 percentage points).

Consistent with these threshold-dependent metrics, the threshold-independent discriminative performance also declines after ablation (Fig. 4b, c). The full TREPP model achieves the best AUROC (0.9598) and AUPRC (0.9471). In contrast, removing gene-related features reduces the AUROC to 0.9375 and AUPRC to 0.8647, while removing sequence-based features further reduces the AUROC to 0.8956 and AUPRC to 0.8195. Overall, these results indicate that both feature groups provide complementary predictive signals. Notably, sequence-derived context features appear to contribute a larger share of the performance gain in our experimental setting, highlighting the importance of modeling local sequence environments.

D. Explaining Model with SHAP

To interpret how TREPP leverages different features to predict TR pathogenicity, we performed SHAP (SHapley Additive exPlanations) analysis. Since TREPP is an ensemble model, we calculated the average SHAP values across all base CatBoost learners. These explanations were computed on the underlying tree models to ensure the results accurately reflect the core decision logic of the ensemble.

Figure 5a displays the global feature importance ranking. Features related to gene structure are the most influential, with `dist_to_exon`, `in_exon`, and `dist_to_cds` emerging as the top predictors. This result is expected, as expansions located within or in close proximity to coding regions are strong candidates for pathogenicity due to their potential impact on protein function or gene expression. However, sequence-based features also play a significant role in the model's decision-making process. Features related to GC content (e.g., `log_tr_gc`) and motif properties (e.g., `delta_gc_tr_motif`) are also highly ranked, indicating that the model relies on a combination of genomic position and intrinsic sequence composition to generate predictions.

The SHAP summary plot (Fig. 5b) further illustrates the directionality of these feature effects. For distance-based features, lower values (representing closer proximity to functional elements) generally correspond to higher SHAP values, confirming that being near functional gene elements increases the predicted risk. Similarly, overlapping with an exon (`in_exon`) strongly contributes to a higher pathogenicity score. Regarding sequence features, higher TR GC content (`log_tr_gc`) is asso-

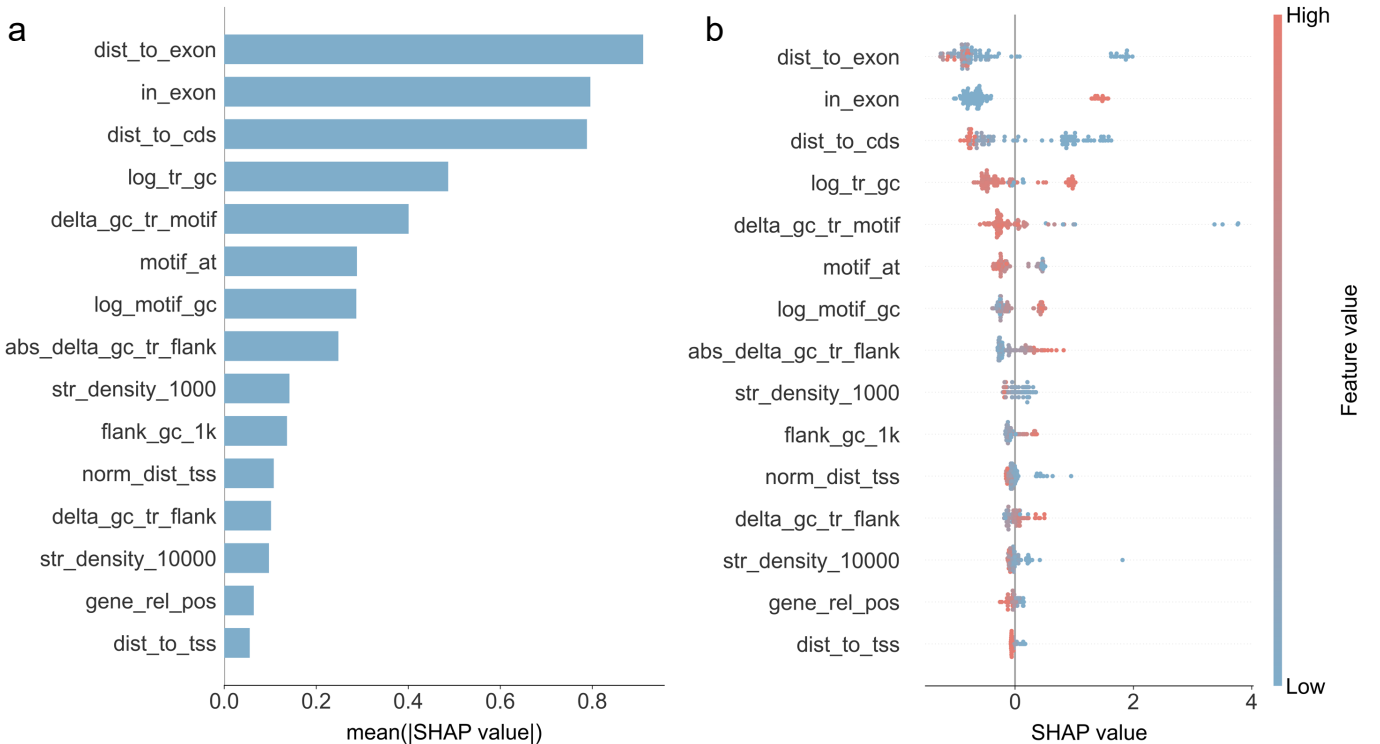


Fig. 5. **SHAP analysis for TREPP features.** (a) Feature importance ranking. (b) Summary plot showing how feature values affect the prediction (red means high feature value, blue means low).

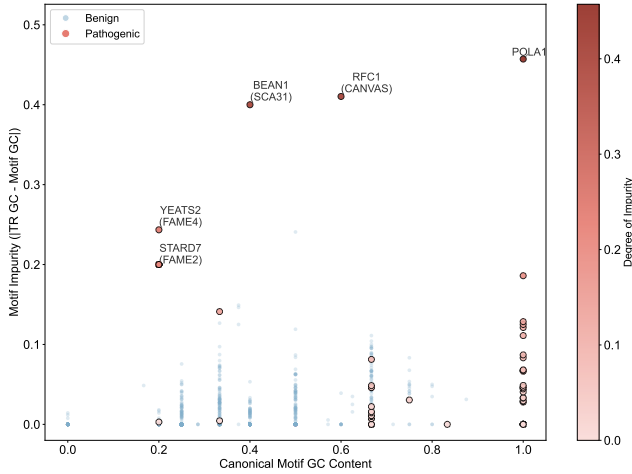


Fig. 6. **Sequence composition landscape reveals distinct pathogenic mechanisms.** The scatter plot delineates benign (blue/grey) and pathogenic (red) loci based on Canonical Motif GC content (x -axis) and Motif Deviation ($|TR\ GC - Motif\ GC|$, y -axis). (A) Benign loci predominantly cluster near zero deviation ($y \approx 0$), indicating sequence stability. (B) Pathogenic loci exhibit significant deviations, capturing two distinct mechanisms: (1) *Pathogenic Motif Interruptions*: Disorders such as CANVAS (*RFC1*) and FAME subtypes show high deviation, reflecting motif insertions that differ from the canonical sequence. (2) *Compositional Instability*: Outliers like *POLA1* exhibit extreme deviation, marking complex genomic instability events.

ciated with increased predicted risk, aligning with previous studies suggesting that GC-rich repeats are often prone to instability [22]–[24]. Additionally, we observed that differences in GC content also contribute to the model’s decision: a larger difference between the TR and its motif ($\Delta_{GC_TR_MOTIF}$) or its flanking region ($\Delta_{GC_TR_FLANK}$) tends to increase the pathogenicity score. This suggests that TRs with GC profiles that are distinct from their immediate genomic context or motif unit are more likely to be prioritized by the model. Finally, the density of other repeats in the surrounding region ($STR_DENSITY_1000$) shows a positive association with the predicted score.

E. TREPP Captures Pathogenic Motif Interruptions and Compositional Instability

Traditionally, TR pathogenicity prediction has focused primarily on repeat tract length and genomic position. However, recent mechanistic studies highlight the critical role of internal sequence composition [23], [25]. Specifically, factors such as motif interruptions and GC bias are now known to drive repeat instability and pathogenicity. To evaluate whether TREPP effectively captures these subtle biophysical signals, we analyzed the relationship between the canonical motif composition and the actual GC content of the expanded tracts (Figure 6).

We observed a distinct divergent distribution that effectively separates pathogenic loci from benign background noise. As shown in Figure 6, benign loci (grey points) cluster tightly around the zero-deviation line, indicating that non-pathogenic repeats tend to maintain high sequence purity consistent with their canonical motifs. In contrast, pathogenic loci (red points) frequently exhibit significant deviations ($\Delta_{GC} > 0.1$). This

separation is driven by two primary mechanisms captured by the feature $\Delta_{GC_TR_MOTIF}$.

First, TREPP successfully identifies “**pathogenic motif interruptions**” in disorders typically associated with AT-rich motifs. Classic examples include Familial Adult Myoclonic Epilepsy (FAME) genes such as *YEATS2* and *STARD7* (Figure 6, left quadrant). In these disorders, the pathogenic mechanism involves the insertion of specific G/C-rich patterns (e.g., TTTTA $\rightarrow$ TTTCA) into typically AT-rich tracts [26]. The model identifies these loci as outliers with high absolute deviations, demonstrating sensitivity to these specific interruption events that simple GC-content features might miss.

Second, the model captures complex compositional instability in high-GC or complex repeats. For instance, *RFC1* (associated with CANVAS) and *BEAN1* (associated with SCA31) appear well outside the benign stability zone. These expansions often involve motif switching (e.g., AAAAG $\rightarrow$ AAGGG in *RFC1*) [27], which dramatically alters the physicochemical properties of the repeat tract. Extreme outliers such as *POLA1* further highlight the model’s capability to detect rare, intronic expansions with unique compositional biases.

Collectively, these results demonstrate that TREPP does not merely bias predictions toward high-GC regions. Instead, the model leverages compositional differences to detect “sequence deviations”. These “deviations” serve as robust markers for genomic instability, enabling precise pathogenicity prediction even for complex or interrupted repeat expansions.

F. Gene Overlap Analysis

We assessed whether top-ranked TR loci map to known disease genes by comparing our predictions to OMIM [28] and GeneTrek [29]. For each method, we selected the top- K loci ($K \in \{500, 1000, 5000, 10000\}$) ranked by pathogenicity score, mapped each locus to the single gene with the largest overlap, and quantified enrichment.

TREPP shows consistent enrichment across multiple cutoffs and databases (Fig. 7). At $K = 500$, TREPP’s top loci map to OMIM genes 374 out of 456 times (82.0%), compared to 293/389 (75.3%) for REXPT. This advantage persists at $K = 10000$, where TREPP achieves 5158/6782 (76.1%) versus REXPT’s 4556/6099 (74.7%). A similar trend is observed for GeneTrek: at $K = 500$, TREPP achieves 89.9% overlap (vs. 85.6% for REXPT), and at $K = 10000$, it maintains 86.1% overlap (vs. 84.9%).

The enrichment advantage holds at both high-stringency ($K = 500$) and lower-stringency ($K = 10000$) cutoffs, indicating robustness across scales. We note that OMIM and GeneTrek are incomplete, so genes not found in these databases are not necessarily false positives but may represent novel candidates. These results suggest that TREPP consistently identifies more known disease genes than REXPT, demonstrating robustness across prioritization scales.

G. Genome-Wide Context and Distribution of High-Risk TR Loci

We next examined whether genome-wide TREPP predictions exhibit interpretable patterns at both the annotation and

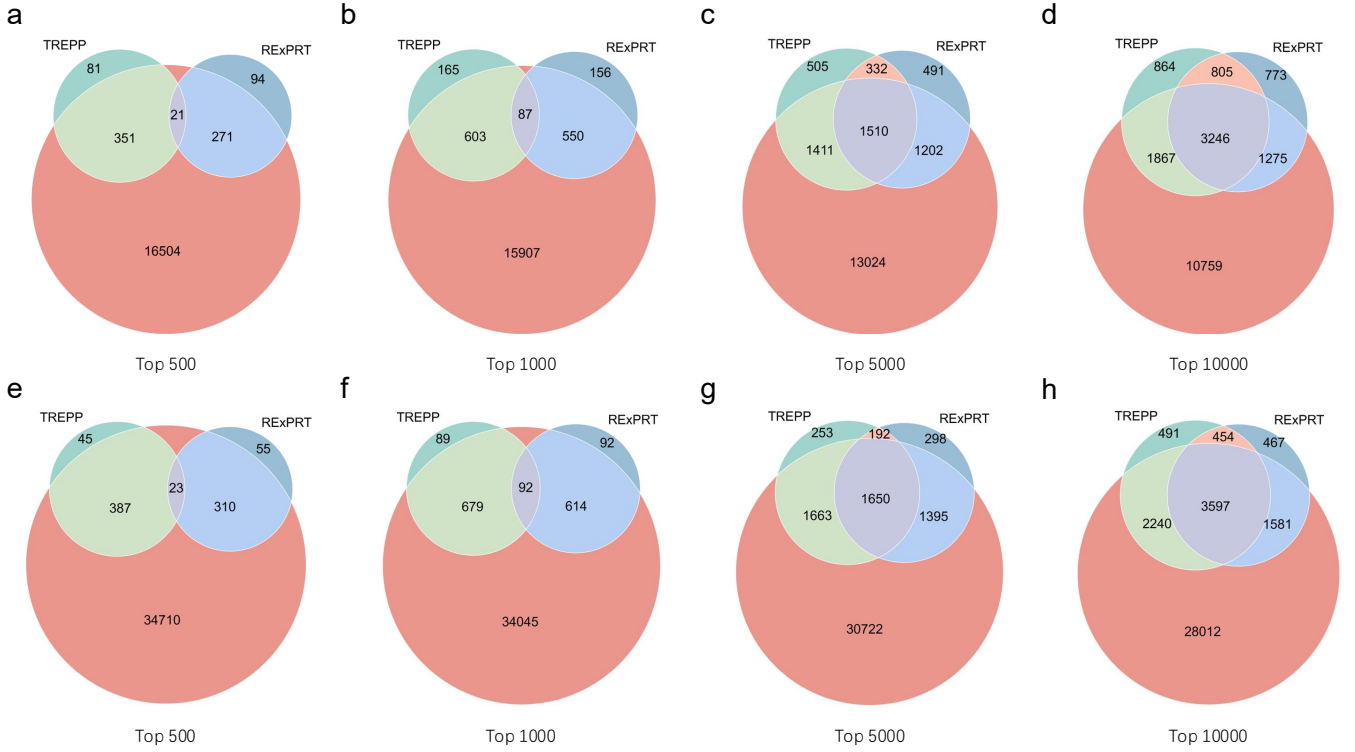


Fig. 7. **Gene overlap at different top- K cutoffs between top-ranked TR loci and curated disease gene resources.** Venn diagrams show the overlap between Ensembl genes mapped from the top- K loci ranked by TREPP or REXPERT and two external gene lists: **OMIM** disease genes (a–d) and **GeneTrek** neurodevelopmental disorder (NDD) genes (e–h). Columns correspond to increasing ranking cutoffs ($K = 500, 1000, 5000, 10000$; left to right). Loci are ranked by predicted pathogenicity scores and mapped to genes using GENCODE gene intervals. To reduce inflation in gene-dense regions, each locus is assigned to a single gene based on the largest coordinate overlap.

chromosomal scales. Across analyses, higher TREPP scores are strongly associated with increased overlap with functionally annotated regions and with localized elevations in the fraction of high-risk calls, even after normalizing for baseline TR candidate density.

High-risk loci are significantly enriched in functional genic annotations (Fig. 8a). Using a matched-background enrichment analysis that controls for motif length, TR length, and TR GC content, high-risk loci (score > 0.5) show substantially higher odds of overlapping exons ($OR \approx 155$) and UTRs ($OR \approx 174$), and moderately higher odds of overlapping promoters ($OR \approx 6.5$). Conversely, high-risk loci are depleted in non-genic contexts, including intronic and intergenic regions. These results indicate that TREPP prioritization is non-random with respect to genome annotation and consistently identifies candidates concentrated in genic functional contexts.

Furthermore, TREPP scores show a strong, monotonic positive association with functional overlap (Fig. 8b). When stratifying loci into score bins spanning the full range of predicted pathogenicity, the percentage of loci overlapping functional regions (defined here as promoters and coding sequences/CDS) exhibits a clear linear increase as the score rises. Unlike simpler threshold-based models, TREPP demonstrates a continuous sensitivity: low-scoring bins (< 0.1) show minimal functional overlap, while the highest-scoring bins (> 0.9) achieve significant enrichment. This relationship is supported by a strong Spearman correlation computed across

bins (Spearman $r = 0.87$, $P = 1.17 \times 10^{-3}$), suggesting that TREPP scores provide a well-calibrated prioritization signal where the score magnitude directly reflects the likelihood of functional importance.

Finally, localized elevations persist after normalizing for TR candidate density (Fig. 8c). To disentangle spatial concentration from heterogeneous TR density, we summarized predictions in 2 Mb windows using the proportion of high-risk loci among all candidates within each window. The resulting map reveals non-uniform patterns across chromosomes, with specific windows exhibiting elevated high-risk ratios. Because this statistic explicitly normalizes by the number of candidates per window, these elevations do not merely reflect baseline TR abundance but instead indicate regional hotspots of putative pathogenicity prioritized by TREPP. Taken together, these findings provide convergent evidence that TREPP prioritization is biologically structured and functionally relevant.

IV. CONCLUSION

Predicting TR expansion pathogenicity is challenging because of the scarcity of validated loci and the complex role of sequence context. Existing tools often overlook repeat-intrinsic properties that influence instability. To address this gap, we introduced TREPP, an interpretable ensemble framework that integrates genomic spatial relationships with sequence-derived features. TREPP uses controlled undersampling and a cal-

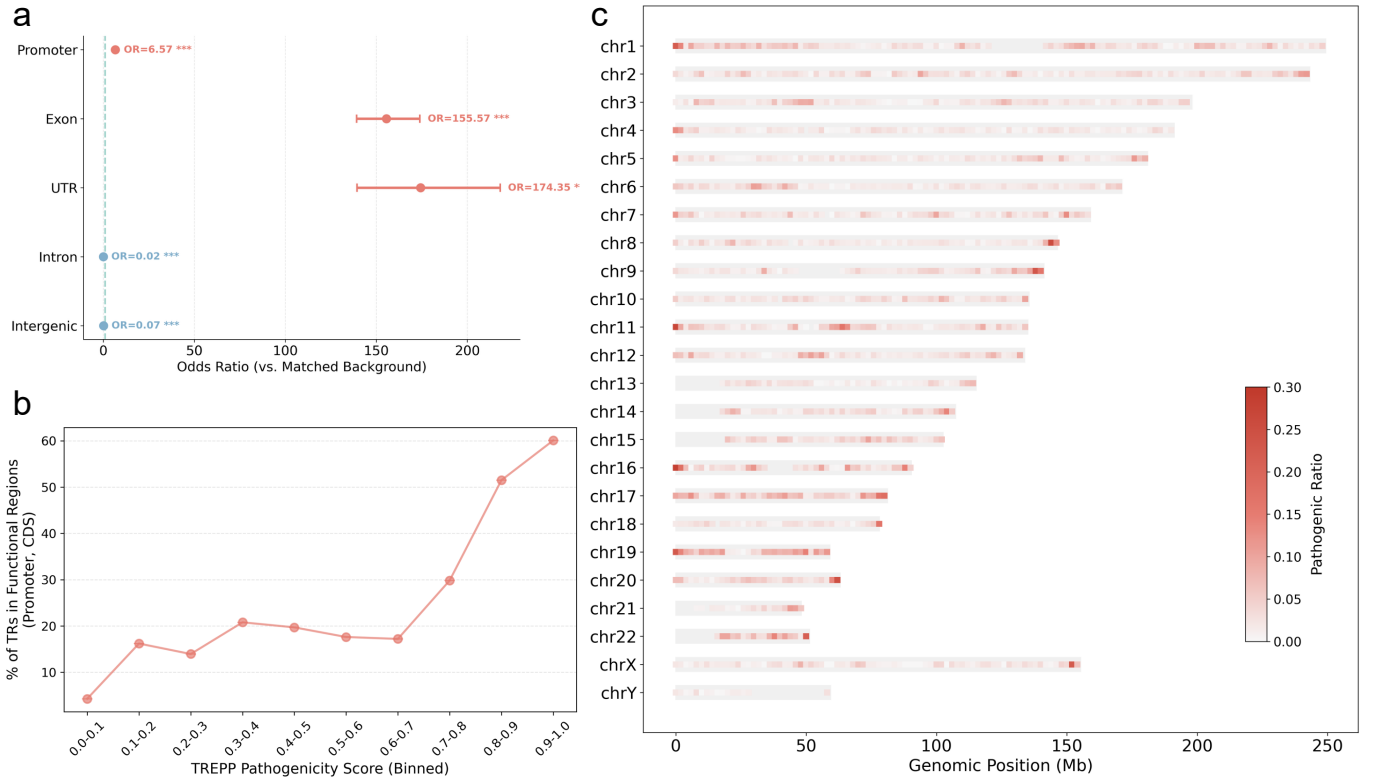


Fig. 8. **Genome-wide contextual analysis of TR loci prioritized by TREPP.** (a) Genomic region enrichment (matched background). High-risk loci (TREPP score > 0.5) are compared against a *matched* background sampled from lower-scoring candidates, controlling for motif length, TR GC content, and TR length. Points show odds ratios (OR) with 95% confidence intervals; significance is assessed by two-sided Fisher's exact test (*ns*, $P \geq 0.05$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$). (b) Score–context association (binned trend). Candidate loci are grouped by predicted score into fixed intervals spanning $[0, 1]$ (width 0.1). For each bin, we report the percentage of loci overlapping functional regions (promoter/CDS). The strong positive trend is quantified by Spearman correlation between the bin mean score and the functional-overlap percentage. (c) Normalized high-risk ratio across the genome. The genome is partitioned into 2 Mb windows; each window is colored by the proportion of high-risk loci among all candidates in that window, thereby normalizing for local TR density. Windows with low candidate counts are omitted to ensure stability.

ibrated stacking meta-learner to handle the extreme class imbalance inherent in TR datasets.

In evaluations, TREPP significantly outperforms RExPRT. Feature analysis confirms that the model leverages biologically relevant signals, such as exon proximity and GC profiles. Notably, by analyzing sequence deviations, we found that TREPP effectively identifies potential pathogenic motif interruptions, capturing mechanisms such as motif switching in disorders like CANVAS and FAME. Furthermore, genome-wide analysis reveals a strong linear enrichment of high-scoring candidates in functional regions and known disease genes, validating the model's biological coherence.

Despite these advancements, limitations remain. First, our model is limited by the small number of validated pathogenic TRs available for training. Second, relying on the reference genome limits the detection of patient-specific repeat variants. Future work will expand the training set, incorporate long-read sequencing data to capture individual-specific repeat structures, and experimentally validate top predictions—moves that will help establish TREPP as a clinically useful screening tool.

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